TRIBHUWAN UNIVERSITY

INSTITUTE OF HUMANITIES AND SOCIAL SCIENCE

MADAN BHANDARI MEMORIAL COLLEGE (MBMC)

ANAMNAGAR, KATHMANDU

A Project Proposal On

College Bus Routing System

Submitted to

DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCE

TRIBHUVAN UNIVERSITY (T.U)

KRITIPUR, KATHMANDU, BAGMATI, NEPAL

In partial fulfillment of the requirements for the degree of Bachelor's in Computer Application (BCA)

Submitted by

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Under the supervision of

Naw Raj Joshi

SUPERVISOR'S RECOMMENDATION

I hereby recommend that this project prepared under my supervision by SRIJANA SUBEDI and ROJ DAHAL entitled **"College Bus Routing System"** in partial fulfillment of the requirements for the degree of Bachelor of Computer Application is recommended for the final evaluation.

……………………………….

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SUPERVISOR

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LETTER OF APPROVAL

This is to certify that this project prepared by Srijana Subedi and Roj Dahal entitled **“College Bus Routing System”** in partial fulfillment of the requirements for the degree of Bachelor in Computer Application has been evaluated. In our opinion it is satisfactory in terms of scope and quality as a project for the required degree.

Evaluation Committee

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ABSTRACT

In the modern era of digitalization, an efficient and well-organized College Bus Routing System is essential for ensuring safe and reliable transportation for students .This project aims to develop a web-based College Bus Routing System that enables effective management of college transportation by providing route information, bus allocation, and real-time Communication.

The system is designed with a responsive and user-friendly front-end using HTML, CSS, and JavaScript, allowing students, drivers, and administrators to access relevant transportation details easily. The backend is developed using PHP and MySQL, which manages user authentication, bus and route management and secure data storage. The system utilizes Dijkstra's Algorithm to calculate the shortest and most efficient bus routes and computes Estimated Arrival Times (ETA) for each stop.

This project documentation also includes system design, architectural overview, database structure, security considerations, and future enhancement possibilities. The College Bus Routing System aims to improve transportation efficiency and provide a reliable and transparent bus service, thereby enhancing the overall campus commuting experience.

Keywords: HTML, CSS, JavaScript, PHP, My SQL

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I am obliged to several people who helped me to organize this project and thankful for their kind opinions, suggestions, and appropriate guidelines. I have received endless support and guidance in finalizing my project so I would like to take this opportunity to thank them all.

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Srijana Subedi

Roj Dahal

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API	:	Application Programming Interface
CSS	:	Cascading Style Sheets
DFD	:	Data Flow Diagram
ER	:	Entity Relationship
ERD	:	Entity Relationship Diagram
ETA	:	Estimated Arrival Times
GPS	:	Global Positioning System
HTML	:	HyperText Markup Language
MY SQL	:	My Structured Query Language
PHP	:	Hypertext Preprocessor
WAN	:	Wide Area Network

CHAPTER 1: INTRODUCTION

1.1 Introduction

Transportation is an essential service in colleges. However, traditional bus management systems often suffer from inefficiencies such as unnecessary stops, poor communication between students and drivers, and inaccurate bus arrival times. To address these challenges, this project proposes a College Bus Routing System**,** a web-based platform designed for students, drivers, and administrators. The system allows students to select their pickup locations and mark daily attendance, while drivers receive information regarding student presence or absence.

The system utilizes Dijkstra's Algorithm to calculate the shortest and most efficient bus routes and computes Estimated Arrival Times (ETA) for each stop. Active pickup points are displayed on a digital map, enabling drivers to avoid unnecessary stops and optimize travel time. By ensuring buses stop only where students are present, the system minimizes delays, improves fuel efficiency, and enhances overall transportation reliability.

Furthermore, the College Bus Routing System enhances safety and convenience by providing students with accurate, real-time bus arrival information. Drivers benefit from timely attendance updates, allowing better route planning and decision-making. Overall, the system ensures smoother and more efficient transportation management by integrating intelligent routing, real-time communication, and monitoring mechanisms.

1.2 Problem Statement

1. **Inefficient Bus Routing:** Buses stop at every location even if no students are present, leading to delays and fuel wastage.

2. **Lack of Student–Driver Communication:** Drivers are unaware of new students or absentees, causing confusion and wasted time waiting for the student.

3. **No Accurate Arrival Time Prediction:** Students do not receive real-time bus arrival information, leading to long waiting times.

1.3 Objectives

1. To develop a web-based system that allows students to select pickup locations, mark attendance, and receive accurate ETA.

2. To help the driver to follow the optimized route provided by the Dijkstra Algorithm .

1.4 Scope and Limitation

1.4.1 Scope

The scope of the College Bus Routing System includes managing college transportation by administrators to manage buses, drivers, and student. Drivers can view the student pick up location, while students can view estimated arrival time. The system focuses on improving transportation coordination, reducing waiting time, and providing accurate route and arrival information within the college transport network.

1.4.2 Limitation

The College Bus Routing System is limited to the college transportation environment and does not cover public or external transport services. Real-time tracking accuracy depends on internet connectivity availability on driver devices. The system does not automatically handle traffic conditions or route changes in real time unless manually updated. Additionally, advanced features such as emergency handling, automated alerts, and integration with external map services are beyond the current system scope.

1.5 Development Methodology

The project will follow the Agile Methodology that will break the project in to phases and will focus on the continuous collaboration and its improvement. A cycle of planning, execution and evaluation will be followed. The following key stages were undertaken:

- Planning and Research: Due to the project's complexity, planning and research were done to figure out the needed tech tools
- Analysis: Requirements for the project were selected, and different tech options

were compared to match our needs.

- Design: During the design phase elements for the project were created based unidentified requirements.
- Implementation: The project was started to make it a working product.
- Testing: The quality and performance of the product were checked and any bugs found during this phase were fixed.

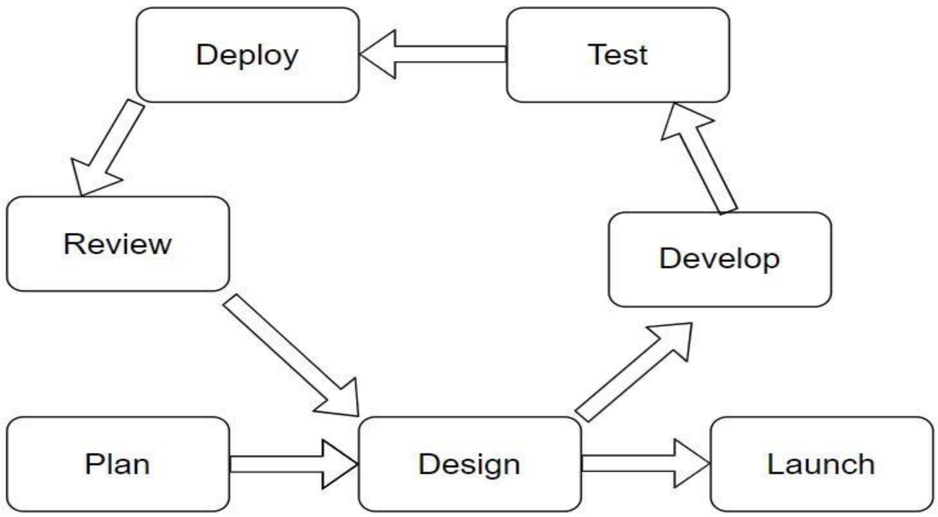

Figure 1.1: Agile Methodology

1.6 Report Organization

The document is organized as follows:

Chapter 1: Introduction, including problem statement, objectives, scope, and methodology.

Chapter 2: Background study and literature review, providing an overview of relevant theories, concepts, and existing studies.

Chapter 3: System analysis , detailing requirements and feasibility analysis.

Chapter 4: System design, structured approach and algorithm details.

Chapter 5: Implementation and testing, details of tool used, implementation of modules, test cases and result analysis.

Chapter 6: Conclusion and future recommendations.

CHAPTER 2:
BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

Most colleges rely on traditional bus systems where buses follow fixed routes and stop at every designated point, regardless of student presence. Attendance is manually tracked or not tracked at all. Drivers are not notified about new students, and students do not know accurate bus arrival times. This makes the existing system inefficient, inconvenient, and time-consuming. There is also no centralized digital platform to manage routes, drivers, or student data, which increases administrative workload. Additionally, the lack of automation leads to frequent delays and inconsistent transportation schedules. Students are also not notified if delays occur.

To address these limitations, the proposed system integrates **Dijkstra's Algorithm**, a graph-based shortest path algorithm used to determine the minimum distance between nodes. In the context of the College Bus Routing System, bus stops are treated as nodes and the distances between them as weighted edges. After processing student attendance data, inactive stops are removed, and Dijkstra's Algorithm is applied to generate the most efficient route covering only active pickup points. This ensures reduced travel distance, optimized fuel usage, and improved operational efficiency.

Furthermore, the system calculates the **Estimated Arrival Time (ETA)** based on the optimized route. ETA is computed by dividing the distance between consecutive stops by the average speed of the bus and accumulating the travel time sequentially. This allows students to receive accurate arrival time notifications while enabling drivers to follow a structured schedule. The integration of route optimization and ETA calculation transforms the traditional transportation system into a dynamic, automated, and intelligent bus management solution.

2.2 Literature Review

Smart Transportation Systems for Educational Institutions: This study analyzes the implementation of intelligent transportation systems within educational institutions, focusing on optimizing college and school bus operations. The authors highlight how route optimization and scheduling systems reduce fuel consumption, travel time, and operational costs while improving student safety and punctuality. The study emphasizes the role of digital platforms in automating transportation management processes. [1]

Shortest Path Algorithms in Public Transportation Routing: This research explores the application of shortest path algorithms, including Dijkstra’s Algorithm, in public and institutional transportation systems. The authors discuss how graph-based routing techniques effectively determine optimal routes by minimizing distance and travel time. The paper demonstrates the suitability of these algorithms for real-time and large-scale transportation networks such as college bus routing systems. [2]

GPS-Based Vehicle Tracking and ETA Prediction Systems: This paper examines the integration of GPS technology with transportation management systems to provide real-time vehicle tracking and estimated arrival time (ETA) prediction. The authors explain how dynamic ETA calculation improves passenger satisfaction by reducing waiting time and enhancing transparency in transportation services. The study is particularly relevant for student bus tracking applications. [3]

Web-Based Transportation Management Systems: This research focuses on the design and development of web-based transportation management systems using modern web technologies. The authors discuss the advantages of using web platforms for managing routes, drivers, schedules, and users in a centralized manner. The study highlights improved accessibility, scalability, and ease of administration for institutions adopting web-based solutions. [4]

In the context of Nepal , Sajha plus is a prominent public transport app in Nepal that provides real-time tracking and ETAs for Sajha Yatayat buses. While it excels at informing commuters of bus locations along fixed routes, it does not support dynamic route changes based on passenger presence. It is Limited to Sajha buses only. .

CHAPTER 3: SYSTEM ANALYSIS AND SYSTEM DESIGN

3.1 System Analysis

The system analysis focuses on the understanding and defining the requirements and feasibility of College Bus Routing System. It ensures that the project meets its objectives with effectiveness and efficiency.

3.1.1 Requirement Analysis

Requirement analysis is an important stage in website development. It involves gathering and analyzing the requirements of the website in order to identify the goals, features, and functions that the system should have.

3.1.1.1 Functional Requirements:

The functional requirements are as follow:

1. Students should be able to register on the system and securely log in to access their profiles.
2. Students should be able to select or update their preferred pickup locations.
3. Students should be able to mark their attendance as present or absent, which will update the system in real-time.
4. Drivers should be able to log in and view only the active stops where students are present.
5. The system should calculate the shortest possible route for bus using Dijkstra’s Algorithm, considering only active stops.
6. The system should calculate and display the Estimated Time of Arrival (ETA) for each active stop, updating dynamically if changes occur.
7. Administrators should be able to manage buses, driver and view pickup stops from a centralized dashboard

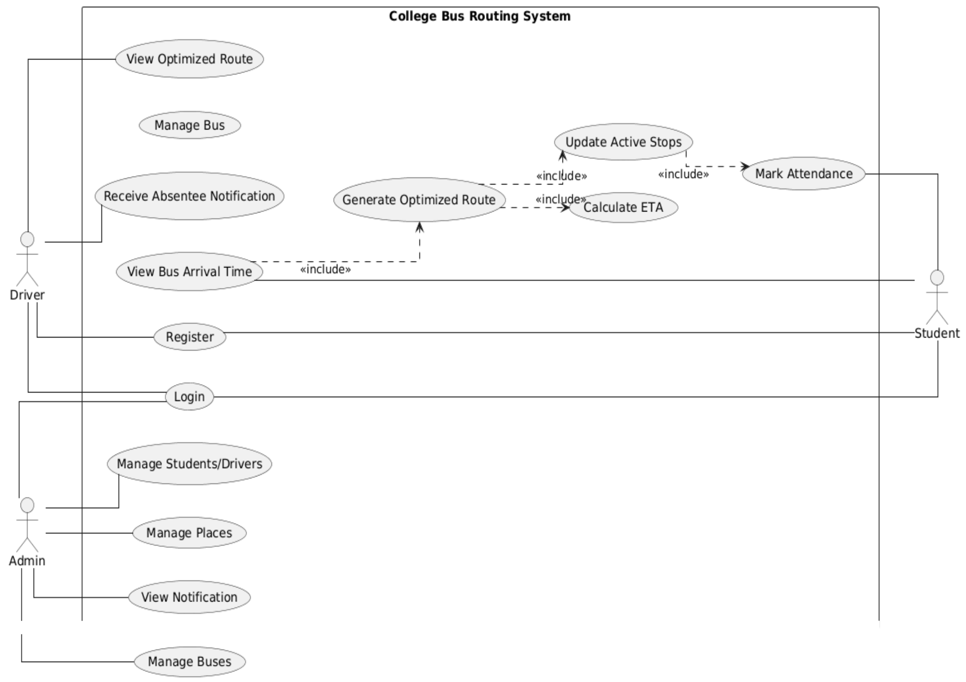

Figure 3.1: Use Case Diagram

3.1.1.2 Non-Functional Requirement

- **Performance**: The system process and analyze user input (including text and data) within 5 seconds on average. It support up to 100 simultaneous users without noticeable performance issues.
- **Scalability**: The system is designed to run locally but must be easily adaptable for cloud or distributed environments, allowing for future expansion by adding more processing power or storage to accommodate increased usage.
- **Security**: User credentials and sensitive data is encrypted. All access and modifications must be logged for auditing purposes, with secure storage to prevent unauthorized access.

- **Usability**: The interface is easy to navigate, with clear instructions for journaling and consultation features. It should be responsive on desktop devices.
- **Reliability**: The system include error handling for issues such as invalid input or failed processing, ensuring graceful recovery without disrupting the user experience.
- **Maintainability**: The system is modular, allowing for easy updates, maintenance, and scalability. The code base is well-documented to facilitate future development and troubleshooting.

3.1.2 Feasibility Study

3.1.2.1 Technical feasibility

The system is technically feasible as the system has a very limited but well optimized services which can be backed up even by limited resources. The system will use HTML, CSS, and JavaScript for the web interface, with PHP for server-side processing and My SQL for database management. Maps will display route and active stops, while Dijkstra's Algorithm will calculate the shortest paths for buses. The system can dynamically update ETA and active stops based on student attendance. All chosen technologies are widely supported, compatible, and suitable for developing an efficient and scalable College Bus Routing System

3.1.2.2 Operational feasibility

Students can mark attendance and select pickup stops easily. Drivers view active stops and follow optimized routes. Admin can manage buses and driver efficiently. The system improves communication between students, drivers, and administrators, reducing confusion and delays. It also allows real-time updates, ensuring that any sudden changes in student attendance or route conditions are handled smoothly.

3.1.2.3 Economic feasibility

This system is economically feasible project uses free or low-cost tools. Fuel and time savings due to optimized routes provide long-term cost benefits. The system also reduces administrative workload, minimizing operational expenses for the college

3.1.2.4 Schedule feasibility

To assure that the project can be successfully accomplished within the allocated timeframe, a comprehensive analysis was conducted, leading to the conclusive determination that the project can indeed be completed comfortably within the estimated timeframe. This analysis involved a thorough examination of the project's scope, resources, and the intricacies of its various phases, ultimately affirming the feasibility of meeting the established timeline without undue constraints or delays.

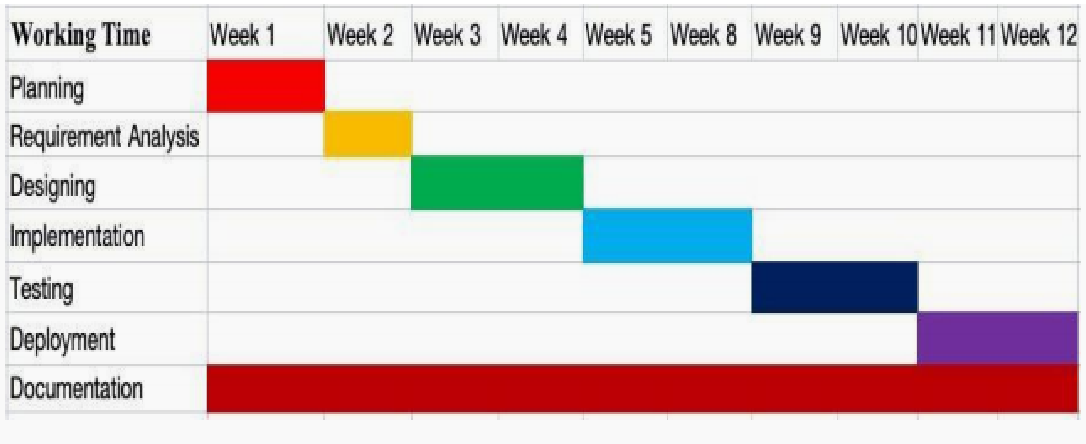

Figure 3.2: Gantt Chart

3.1.3 Analysis

3.1.3.1 Data modeling using ER Diagrams

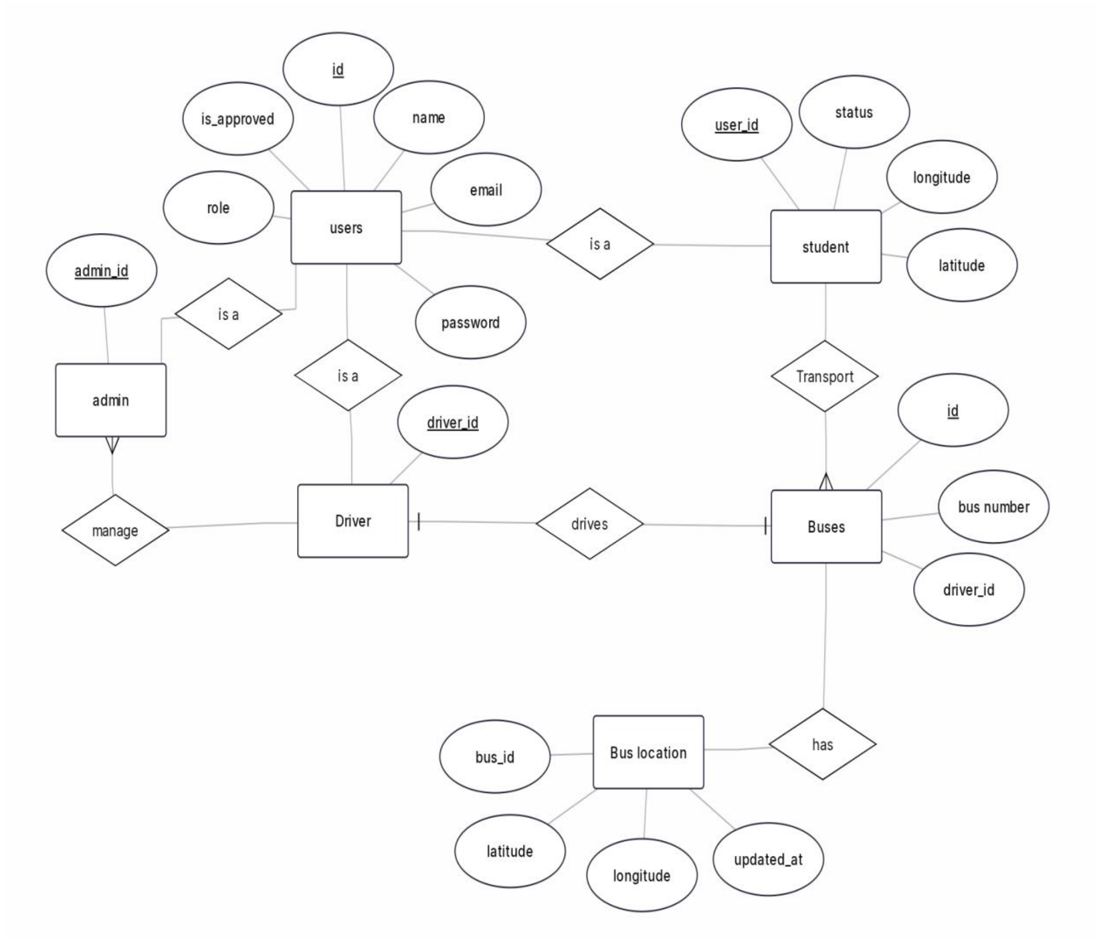

Figure 3.3: ER diagram

This ER diagram represents the **College Bus Routing System**, showing the relationships between Users, Admin, Students, Drivers, Buses, and Bus Location. Users register with basic details such as *user ID, name, email, password,* and *role*. Based on their role, users act as Admin, Students, or Drivers. Admin manage buses and drivers, while drivers operate assigned buses. Each bus is linked to a bus location entity that stores real-time *latitude, longitude,* and *update time*. The system efficiently manages users and bus operations.

3.1.4 Process modeling using DFD

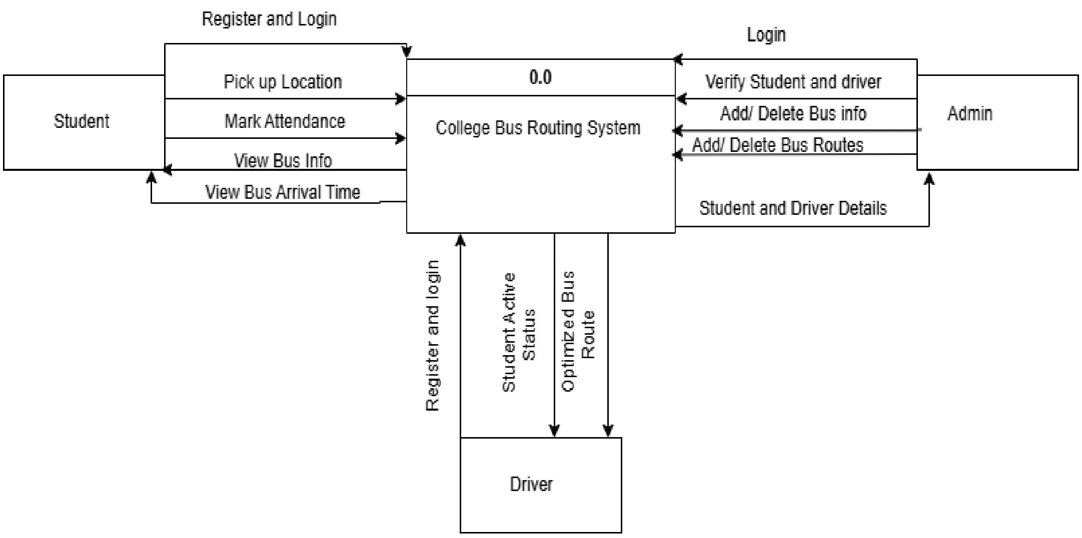

Figure 3.4: DFD Level 0 of College Bus Routing System

This Level 0 DFD illustrates the interaction flow in the **College Bus Routing System**, involving three primary roles: Admin, Driver, and Student. The Admin manages buses, students, and drivers, ensuring proper system operation. Students access the system to view estimated arrival time. The system processes user inputs and provides route and arrival information, enabling efficient communication between students, drivers, and administrators for smooth transportation management.

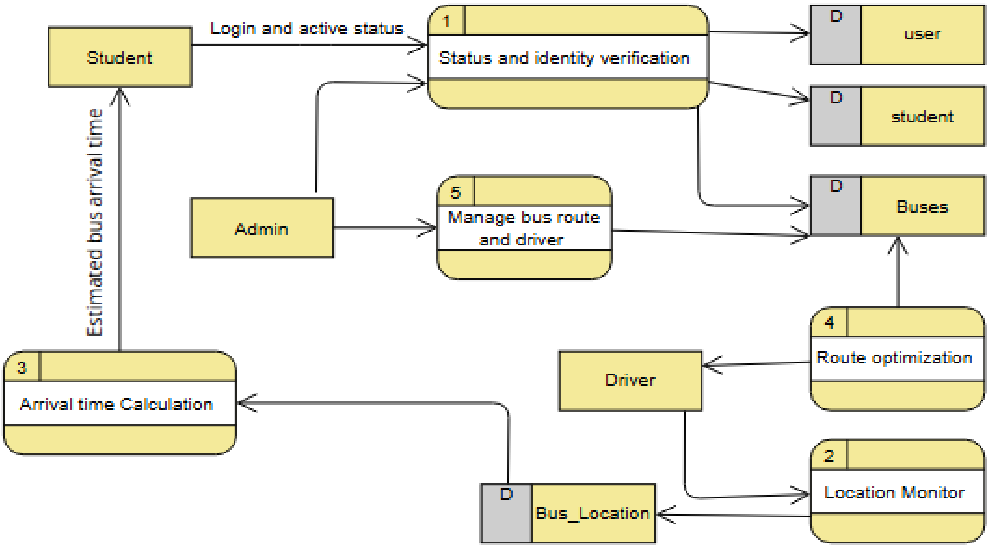

Figure 3.5: DFD Level 1of College Bus Routing System

This Level 1 DFD illustrates the interaction flow in the College Bus Routing System, involving three primary roles: Admin, Driver, and Student, and key processes such as Status and Identity Verification, Location Monitoring, Route Optimization and Arrival Time Calculation. The Admin manages drivers, ensuring smooth system operation and accurate data maintenance. Students access the system through the status and identity verification process and receive estimated bus arrival times calculated by the system. The system performs route optimization based on available data and supports arrival time calculation for students. Through coordinated data processing and interaction among users, processes, and data stores, the system ensures efficient bus routing , accurate arrival predictions, and structured transportation management.

SYSTEM DESIGN

3.2 Design

3.2.1 Architectural design

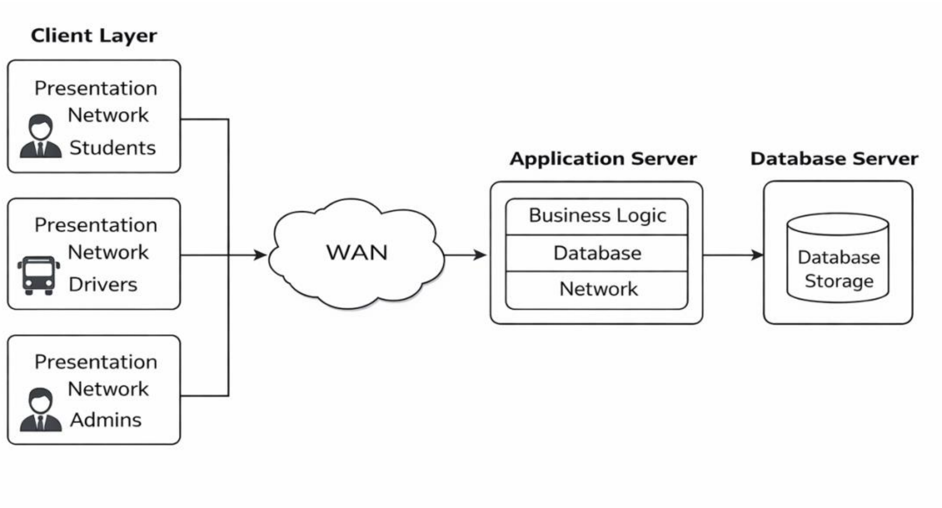

Figure 3.6: Architecture Design of College Bus Routing System

The College Bus Routing System is developed with a focus on simplicity, reliability, and efficient transportation management, adopting a three-tier architecture to organize the system effectively. This architecture is divided into the presentation layer, business logic layer, and data access layer, each responsible for specific functions. The presentation layer serves as the front-end interface for students, drivers, and administrators, providing a user-friendly and responsive design for route viewing and system management. The business logic layer handles core system operations, including user authentication,route optimization, and arrival time calculation. The data access layer interacts with the underlying database to securely store and retrieve information related to users, buses, drivers and location updates. This layered architecture enhances scalability, maintainability, and security, ensuring a reliable and efficient college transportation system for all users.

3.2.2 Database Design: Relational Schema Diagram

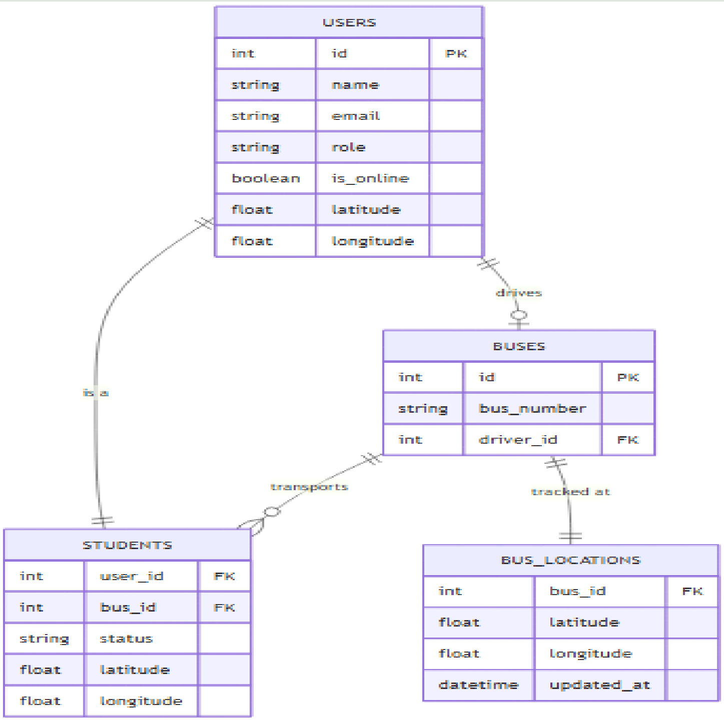

Figure 3.7 : Database Design

The diagram consists of four entities: **Users, Student, Buses, and Bus Location**. The Users table stores common user details, while Admin, Student, and Driver represent role-based user types. The Buses table maintains bus information and driver assignments. The Bus Location table stores real-time location data. The relationships illustrate user roles and bus–driver associations. This design supports efficient bus routing and structured access for all users.

3.2.3 Interface designs

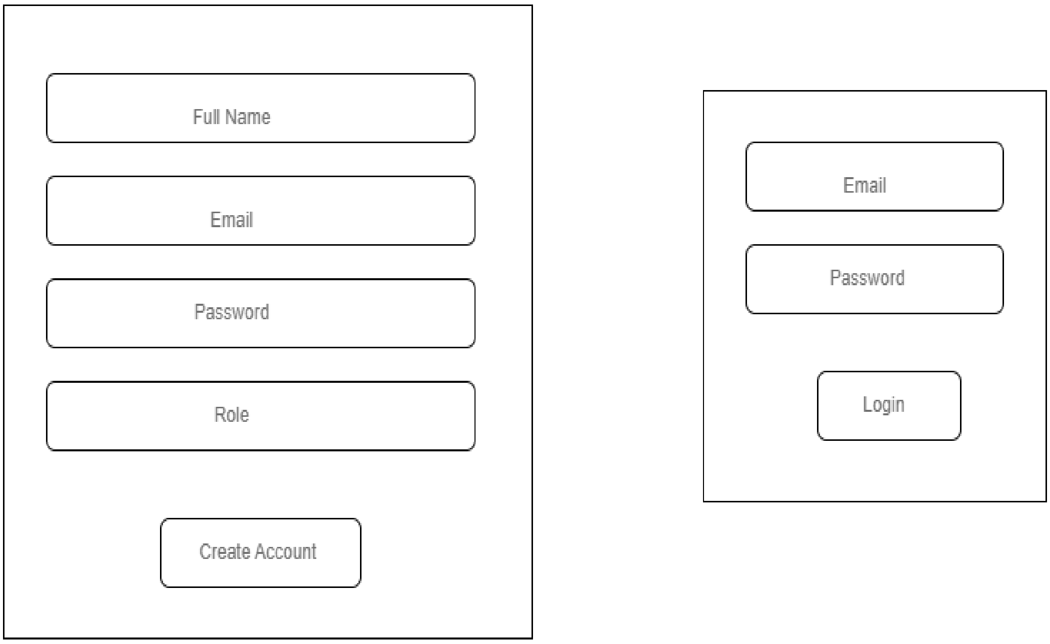

Figure 3.8: Interface design for registration and login

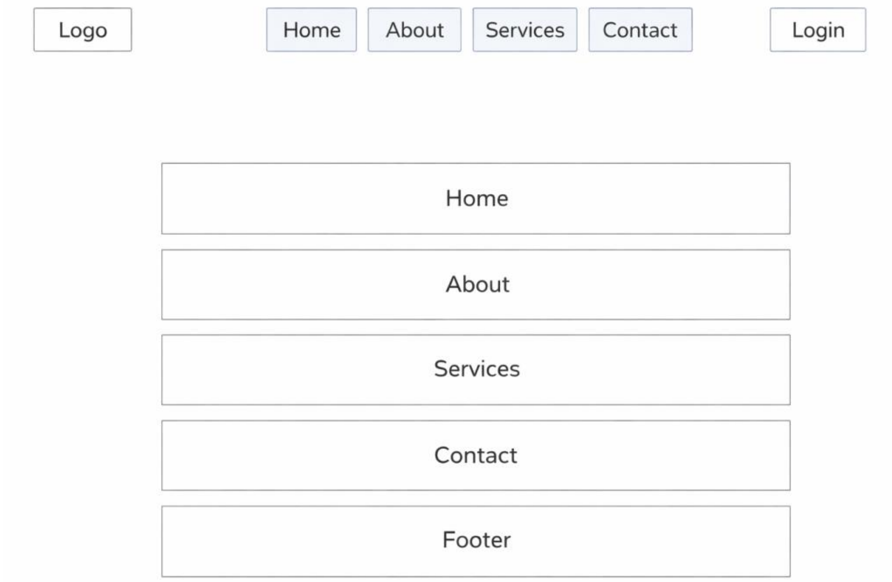

Figure 3.9: Interface design for Landing page

3.2.4 Physical DFD

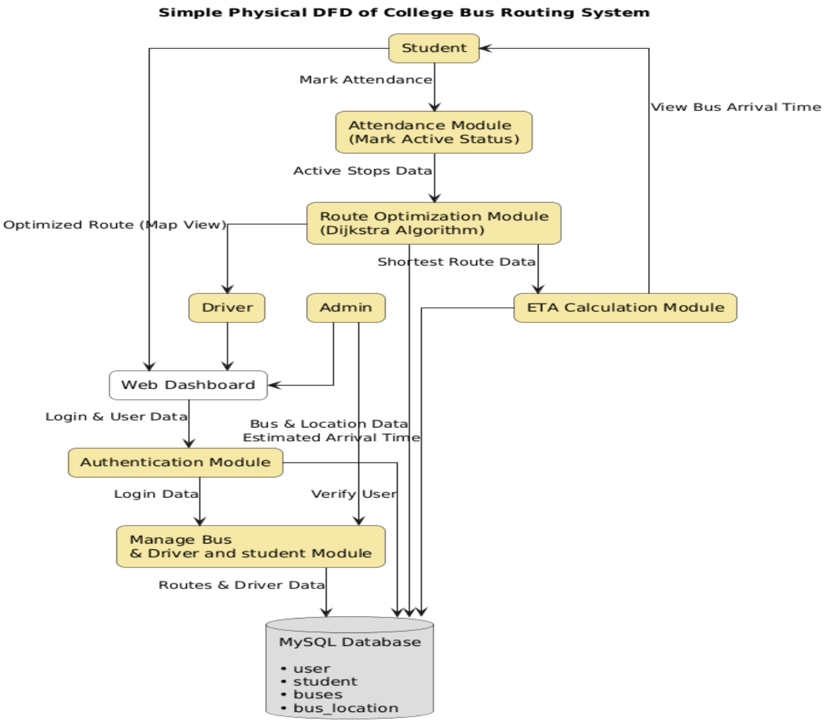

Figure 3.10: Physical DFD

The Physical DFD consists of key components: Student, Driver, Admin, Web Dashboard, Authentication Module, Attendance Module, Route Optimization Module, ETA Calculation Module, and MySQL Database. The Student marks attendance through the Attendance Module, which updates active stops data. This data is processed by the Route Optimization Module using Dijkstra's Algorithm to generate the shortest route. The ETA Calculation Module computes the estimated arrival time based on the optimized route. The Authentication Module verifies user login and controls access. All system data are stored in the MySQL Database. The overall flow ensures efficient route planning, accurate arrival time prediction, and structured management of the college transportation system

3.3 Description of Algorithms

3.3.1 Dijkstra Algorithm

Dijkstra's Algorithm is a graph-based shortest path algorithm used to determine the minimum distance or minimum-time route between a source node and destination nodes in a weighted graph. In the College Bus Routing System, this algorithm is applied to identify the most efficient bus routes between the college and multiple pickup or drop-off locations by minimizing travel distance or estimated travel time.

The transportation network is represented as a graph where bus stops are modeled as nodes and the roads connecting them are represented as edges with associated weights.

1. **Graph Representation**

The routing system models the road network as a weighted graph:

- Vertices (Nodes): Bus stops, pickup points, and the college destination
- Edges: Roads connecting the bus stops
- Weights: Distance or estimated travel time between bus stops

This representation allows the algorithm to efficiently process real-world routing data and compute optimal paths.

2. **Initialization**

At the start of the algorithm:

- The distance from the source node to itself is initialized as 0.
- All other nodes are assigned an initial distance value of infinity (∞).
- A priority queue is used to always select the node with the minimum tentative distance.

Mathematical representation:

dist (source) = 0

dist(other nodes) = ∞

3. **Shortest Path Calculation**

The algorithm iteratively performs the following steps:

- Select the unvisited node with the smallest tentative distance.
- Mark the selected node as visited.
- Update the distances of all adjacent nodes using the formula:

dist(v) = min(dist(v), dist(u) + weight(u, v))

Where:

- dist(u) represents the shortest known distance to node *u*
- weight(u, v) represents the distance or travel time between nodes *u* and *v*
- dist(v) is the updated distance to node *v*

This process continues until the destination node (college) is reached or all nodes have been processed.

4. **Optimal Route Selection**

   After execution, the algorithm produces:

   - The shortest path from the source to the destination
   - An ordered sequence of bus stops along the route
   - The minimum total distance or travel time

3.3.2 Estimated Arrival Time (ETA) Calculation

Estimated Arrival Time (ETA) calculation is used to predict the expected arrival of the bus at each pickup point and at the college. The ETA is computed based on the shortest route obtained from Dijkstra's Algorithm.

1. **Distance-Based ETA Calculation**

   ETA is calculated using the formula:

   ETA = Total Distance / Average Bus Speed

   Where:

   - Total Distance: Sum of edge weights along the shortest route
   - Average Bus Speed: Predefined average speed of the bus (e.g., km/h)

2. **Stop-wise ETA Calculation**

   For each bus stop along the route, ETA is calculated incrementally:

   ETA $(stop_i)$ = ETA(previous stop) + Travel Time$(stop_{i-1} \rightarrow stop_i)$

   This enables the system to display estimated arrival times for each pickup location.

3. **Output Generation**

   The final ETA and route information is displayed to:

- Students: To view bus arrival time and reduce waiting time
- Drivers: To follow optimized schedules
- Administrators: For monitoring and transportation planning.

CHAPTER4: IMPLEMENTATION AND TESTING

4.1 Implementation

4.1.1 Tools Used

- **Visual Studio Code**: It was used as a test editor which helped to run programming, which is used as the major programming language in our system. It comes with built-in support for HTML , and CSS and has a rich ecosystem of extensions for other languages and runtimes.
- **HTML**: It is used in our system to make frameworks as it implements a markup- based pattern to define the behavior of content.
- **CSS**: It is used to describe the presentation of documents made in HTML format.
- **JavaScript**: It was used for building both client-side and server-side scripting in the project, enabling dynamic functionality and interactive features on the web application.
- **React**: It was utilized for developing the client-side scripting of the project, providing a component-based approach to building user interfaces efficiently.
- **Node**: It was implemented for server-side scripting, handling backend operations and managing server functionality for the project.

4.1.2 Implementation Details of modules

1. System Setup and Initialization

Purpose:

Initializes the College Bus Routing System by loading bus stops, distance information, and required system parameters. This setup prepares the system for route optimization and ETA calculation.

Inputs:

- bus_stops (list): A list of all pickup and drop locations.
- distance_matrix (2D array): Represents the distance or travel time between each pair of bus stops.

Process:

The system stores the bus stop data and distance matrix. These values are later used by the routing algorithm to compute the shortest path. Initial data structures for visited stops, shortest distances, and route tracking are prepared.

2. Attendance Processing Module

Purpose:

Processes student attendance data to identify present and absent students and determine active pickup stops for the current day.

Input:

- attendance_data : Contains student IDs mapped to their attendance status (present/absent).

Output:

Active stops (list): A list of pickup stops where at least one student is present.

Process:

The system checks attendance records and removes pickup stops associated only with absent students. The updated list of active stops is forwarded to the route optimization module.

3. Route Optimization Module

Purpose:

Generates the shortest and most efficient bus route covering all active stops using Dijkstra's Algorithm**.**

Input:

- source_stop (int/string**):** The starting location of the bus (e.g college).

Output:

optimized_route (list): Ordered list of bus stops representing the shortest route.

Process:

The system applies Dijkstra's Algorithm to compute the minimum distance from the source stop to all other active stops. The shortest paths are calculated iteratively, selecting the nearest unvisited stop until all required stops are covered.

4. Estimated Time of Arrival (ETA) Calculation

Purpose:

Automatically calculates the Estimated Arrival Time (ETA) for each active bus stop based on the optimized route.

Input:

- optimized_route (list): The route generated by Dijkstra's Algorithm.
- average_speed (float): Average speed of the bus (km/h).

Output:

- eta_details : Maps each bus stop to its estimated arrival time.

Process:

The system calculates travel time between consecutive stops using distance and speed. These times are accumulated to generate ETA values, which are then stored and displayed to students.

4.2 Testing

4.2.2 Test cases

Table 4.1: Testing for login

Test Case Description	Email	Password	Expected Result	Actual Result
User login with correct credentials	Student@gmail.com	112233	Redirected to the Home page	Successfully redirected to the Home page
User login with incorrect credentials	123@gmail.com	Invalid user	Unsuccess full login message	Received an “Email doesn’t exist in our system” message

Table 4.2: Testing for admin login

Test Case ID	Test Case Description	Email	Password	Expected Result	Actual Result
TC001	Admin login with correct credentials	admin123@gmail.com	12345	Redirected to the Dashboard	Successfully redirected to Dashboard
TC002	Admin login with incorrect credentials	adddmin@gmail.com	Invalid admin	Unsuccessful login message	Received an "Email doesn’t exist in our system" message

Table 4.3: Testing for registration

Test Case ID	Test Case Description	Email	Password	Expected Result	Actual Result
TC001	User register with correct credentials	Valid email	valid pass	Waiting for the admin to approve.	Waiting for the admin to approve.
TC002	User register with incorrect credentials	invalid email	invalid pass	Unsuccessful login message	Received an"Invalid credentials " message

Table 4.4: Testing for Student login

Test Case ID	Test Case Description	Email	Password	Expected Result	Actual Result
TC001	User Login with correct credentials	Student@gmail.com	112233	Redirected to the Dashboard	Successfully redirected to Dashboard
TC002	User login with incorrect credentials	sttutee@gmail.com	St6y	Unsuccessful login message	Received an invalid email .

Table 4.5: Testing for System

Test Case ID	Test Case Description	Actions/Input Data	Expected Result	Actual Result
TC001	Active Stop Visualization	Student clicks "Present" on their mobile app.	The student's location appears on the map as an Active stops.	Map icon updates to show "Present" status.
TC002	Absentee Route Exclusion	Student toggles status to "Absent" .	The stop icon is removed from the map .	Stop disappears from the driver's map view.
TC003	Shortest Path Calculation	Driver start from College.	System identifies all "Present" students and uses Dijkstra's Algorithm for shortest route.	Successfull route generation.

4.3 Result Analysis

The College Bus Routing System efficiently processes student attendance data, bus stop information, and distance matrices to generate optimized bus routes and estimated arrival times (ETA). The outcomes demonstrate improved route efficiency, reduced travel time, and better communication between students and drivers.

4.3.1 Route Optimization Analysis

The route optimization analysis evaluates the effectiveness of the system in generating the shortest and most efficient bus route using Dijkstra's Algorithm. When students mark their attendance, the system dynamically updates active pickup stops by removing locations where all students are absent. This reduces unnecessary stops and travel distance.

Testing results show that the optimized routes are consistently shorter than fixed routes, leading to reduced fuel consumption and travel time. The algorithm successfully identifies the minimum-distance path covering all active stops, proving the effectiveness of the routing model.

4.3.2 ETA Accuracy Analysis

The ETA values are calculated automatically based on the optimized route, distance between stops, and average bus speed. The system provides reliable ETA information to students, improving punctuality and reducing waiting time at bus stops.

4.3.3 Attendance-Based Stop Reduction Analysis

This analysis examines the impact of student attendance on route efficiency. When students are marked absent, the corresponding pickup stops are excluded from the route. Testing shows that skipping inactive stops significantly reduces route length and travel duration.

Drivers also receive notify about absent students, enabling them to follow the updated route without confusion. This feature improves operational efficiency and ensures smoother bus operations.

CHAPTER 5: CONCLUSION AND FUTURE RECOMMENDATIONS

5.1 Conclusion

The College Bus Routing System successfully achieves its objectives by providing an efficient and intelligent solution for managing student transportation. By integrating student attendance data with route optimization using Dijkstra's Algorithm, the system dynamically generates the shortest and most efficient bus routes while eliminating unnecessary stops caused by absent students. This approach significantly reduces travel time, fuel consumption, and operational inefficiencies.

The system calculates estimated arrival times (ETA) based on optimized routes and provides real-time information to students, helping them plan their schedules more effectively. At the same time, drivers receive timely notifications about absent students and updated routes, enabling smoother and more organized bus operations. The inclusion of role-based access for administrators ensures proper management of students, bus stops, enhancing overall system reliability.

With a user-friendly interface and data-driven routing decisions, the College Bus Routing System improves communication between students, drivers, and administrators.

5.2 Lesson learn / Outcomes

The College Bus Routing System successfully proved that using Dijkstra's Algorithm makes student transportation much faster and smarter. A major outcome of this project was the ability to change the bus route in real-time; when a student marks themselves as "Absent," the system automatically skips that stop and recalculates a shorter path. This taught us that data-driven routing significantly reduces wasted fuel and travel time compared to traditional fixed routes.

Another key lesson was the importance of real-time communication. We learned that providing live Estimated Arrival Times (ETA) helps students manage their time better and reduces the stress of waiting. By connecting the driver's navigation directly to the students' "Present" or "Absent" status, we created a synchronized system that improves the daily commute for everyone involved.

5.3 Future Recommendations

Although the College Bus Routing System provides an efficient solution for optimizing bus routes using Dijkstra’s Algorithm and attendance-based stop management, several enhancements can be incorporated in the future to further improve system performance and usability.

1. Real-Time Traffic Integration
   Future versions of the system can integrate real-time traffic data using GPS and traffic APIs. This would allow dynamic route adjustments in case of congestion, accidents, or road closures, resulting in more accurate ETA predictions.
2. Mobile Application Development
   A dedicated mobile application for students and drivers can be developed to provide push notifications, real-time bus tracking, and instant attendance updates, improving accessibility and user experience.
3. GPS-Based Live BusTracking
   Integrating GPS devices in buses would enable live tracking of bus locations. Students could view real-time bus movement on a map, increasing transparency and reducing waiting time at stops.
4. Advanced Route Optimization Algorithms
   In the future, advanced algorithms such as A* or machine learning-based route prediction models can be implemented to further enhance routing efficiency, especially for large-scale transportation systems.
5. Scalability for Multiple Campuses
   The system can be enhanced to support multiple campuses, multiple buses, and different schedules, making it suitable for large educational institutions.

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[2] E. W. Dijkstra, “A note on two problems in connexion with graphs,” *Numerische Mathematik*, vol. 1, no. 1, pp. 269–271, Dec. 1959.

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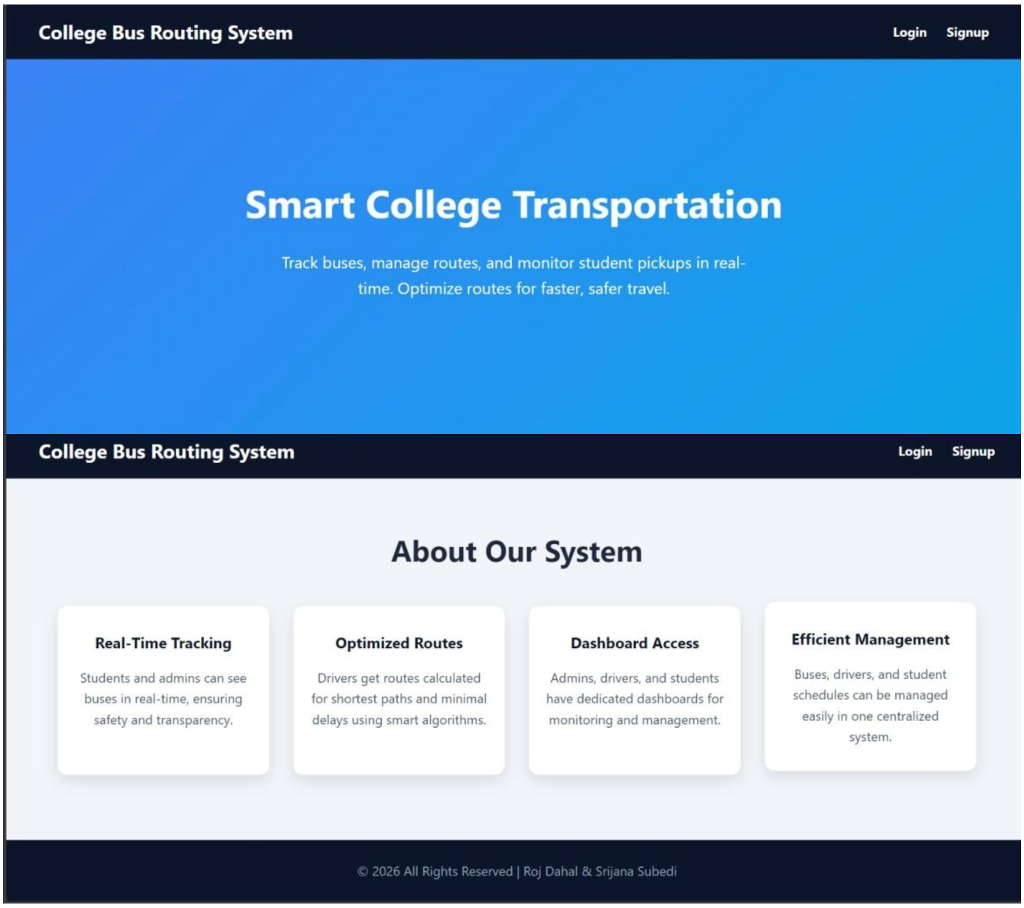
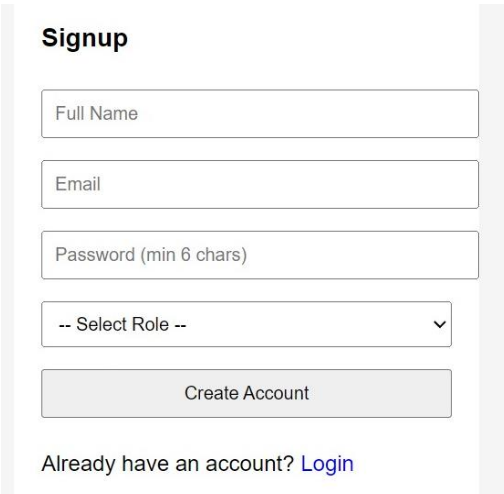
[4] A. Singh and S. Mishra, “Web-based transportation management system,” *International Journal of Computer Applications*, vol. 121, no. 18, pp. 22–26, July 2015

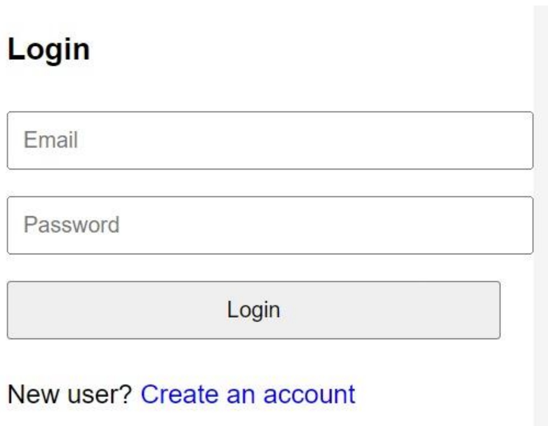
APPENDIX

Landing pages

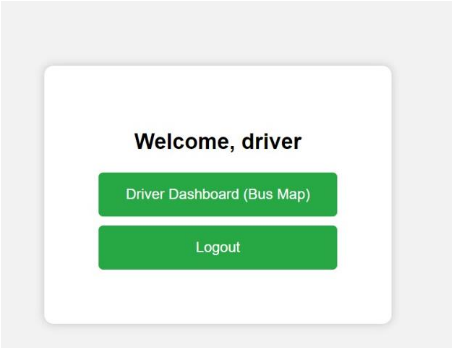


Registration and login page

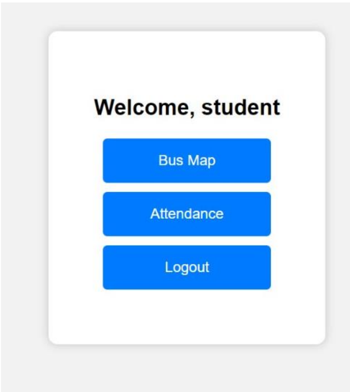


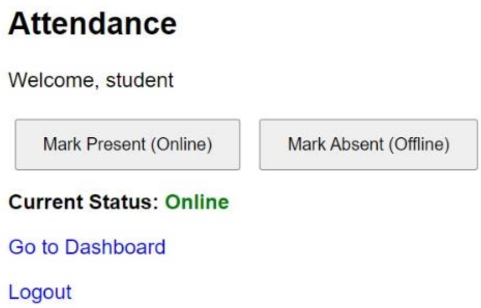


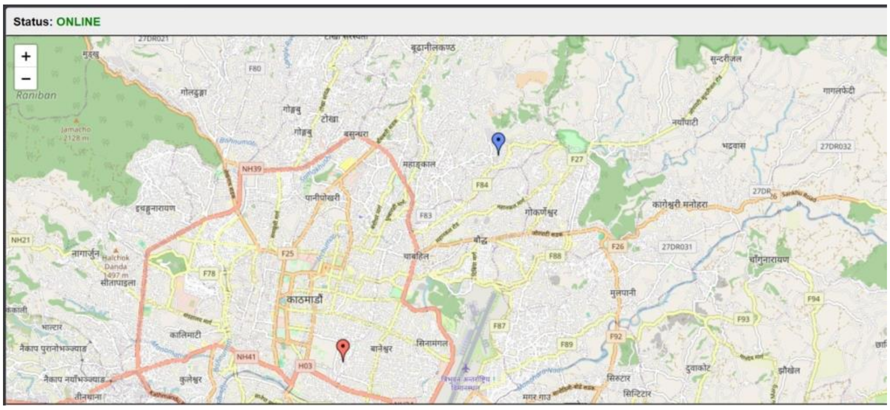
Driver Site



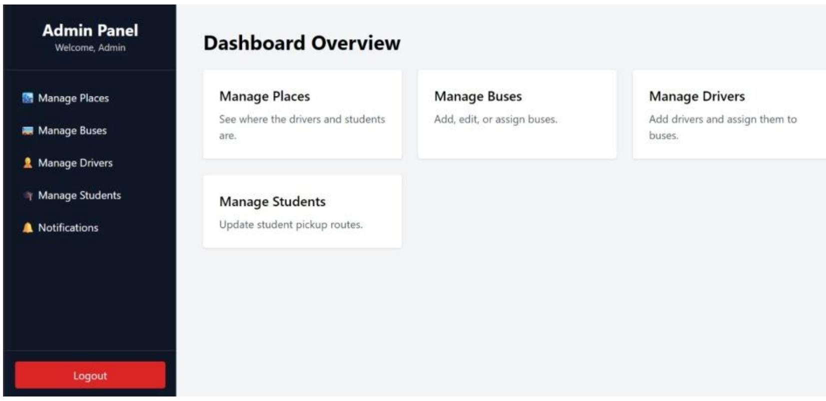
Student site







Admin Site


Admin Panel
Welcome, Admin
Manage Places
Manage Buses
Manage Drivers
Manage Students
Notifications
Logout
Dashboard Overview
Manage Places
See where the drivers and students are.
Manage Buses
Add, edit, or assign buses.
Manage Drivers
Add drivers and assign them to buses.
Manage Students
Update student pickup routes.


Manage Drivers
Add Driver
Name:
Email:
Password:
Add Driver
ID
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Email
Latitude
Longitude
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driver
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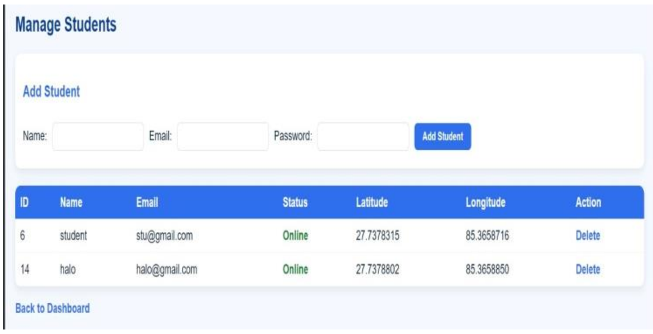
Delete
Back to Dashboard


Manage Students
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